

AI-Powered Chest Disease Detection using Deep Learning, Explainable AI, and Retrieval-Augmented Generation

Project Report

Student Project: NIH ChestXray14 Disease Detection

1. Introduction

Chest X-rays are among the most frequently used medical imaging techniques for diagnosing lung and heart-related diseases. However, interpreting large volumes of radiographic images can be time-consuming and may lead to diagnostic variability.

The objective of this project is to develop an Artificial Intelligence (AI)-assisted disease detection system capable of automatically identifying multiple thoracic diseases from chest X-ray images. The system utilizes deep learning models, Explainable AI (Grad-CAM), and Retrieval-Augmented Generation (RAG) to provide both predictions and clinical explanations.

This project uses the NIH ChestXray14 dataset and compares three different deep learning architectures:

- Custom Convolutional Neural Network (CustomCNN)
- ResNet18
- VGG19

The project focuses not only on disease detection accuracy but also on model interpretability and clinical usability.

2. Problem Statement

Manual interpretation of chest radiographs requires significant expertise and can be challenging when handling large volumes of patient data.

The goal of this project is to:

- Detect multiple diseases from chest X-ray images.
 - Compare different deep learning architectures.
 - Improve model transparency using visual explanations.
 - Generate clinically meaningful summaries using a medical knowledge base.
 - Prevent data leakage through patient-wise data splitting.
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3. Dataset Description

Dataset

NIH ChestXray14 Dataset

Dataset Statistics

- More than 112,000 chest X-ray images
- More than 30,000 patients
- Multi-label classification problem
- 14 disease categories

Disease Labels

1. Atelectasis
2. Cardiomegaly
3. Effusion
4. Infiltration
5. Mass
6. Nodule
7. Pneumonia
8. Pneumothorax
9. Consolidation
10. Edema
11. Emphysema
12. Fibrosis
13. Pleural Thickening
14. Hernia

4. System Architecture

The complete workflow of the proposed system is shown below.

Chest X-Ray Image

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Data Preprocessing

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Patient-wise Train/Test Split

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Deep Learning Models

(CustomCNN / ResNet18 / VGG19)

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Disease Prediction

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Grad-CAM Visualization

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Clinical Knowledge Retrieval

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AI Clinical Summary Generation

↓

Physician Review

Screenshot 1

Insert the Overall Project Workflow Diagram.

5. Data Preparation

Data preparation was performed before model training.

Steps Performed

- Loaded NIH metadata file (Data_Entry_2017.csv)
- Removed "No Finding" labels
- Converted labels into binary multi-label format
- Applied image normalization
- Applied image augmentation techniques

Data Augmentation

Training images were augmented using:

- Random Horizontal Flip
- Random Rotation
- Image Resize (224×224)

These augmentations improve model generalization and reduce overfitting.

Screenshot 2

Insert Dataset Loading Output.

Example:

Using device: cuda

Working dataset size: 10000

6. Patient-wise Data Split

A patient-wise split strategy was implemented to avoid data leakage.

Why Patient-wise Split?

If images belonging to the same patient appear in both training and testing datasets, the model may memorize patient-specific patterns rather than learning disease-specific features.

This leads to artificially inflated performance.

To prevent this issue:

- Patients were separated before splitting.
- Training and testing sets contain completely different patients.

Leakage Verification

Patient overlap was explicitly checked.

Result:

Patient overlap = 0

Leakage-safe split confirmed.

Screenshot 3

Insert Leakage Verification Output.

7. Deep Learning Models

Three architectures were evaluated.

7.1 CustomCNN

CustomCNN is a lightweight architecture built specifically for this project.

Architecture

Input Image

↓

Conv2D

↓

ReLU

↓

Max Pooling

↓

Conv2D

↓

ReLU

↓

Max Pooling

↓

Fully Connected Layers

↓

14 Disease Outputs

Advantages

- Fast training
- Lower computational requirements
- Easy to understand

7.2 ResNet18

ResNet18 is a transfer learning architecture utilizing residual connections.

Advantages

- Pretrained on ImageNet
- Better feature extraction
- Faster convergence

The final classification layer was modified to produce 14 outputs.

7.3 VGG19

VGG19 is a deep convolutional neural network known for strong feature extraction capabilities.

Advantages

- Deep architecture
- High representation capacity
- Strong performance in image classification tasks

The final classifier was modified to predict 14 disease classes.

Screenshot 4

Insert Model Architecture Screenshot.

8. Handling Class Imbalance

The NIH ChestXray14 dataset is highly imbalanced.

Certain diseases occur much less frequently than others.

Examples:

- Hernia
- Fibrosis
- Emphysema

occur much less frequently than:

- Infiltration
- Effusion

To address this issue, weighted Binary Cross Entropy Loss was used.

Loss Function

`BCEWithLogitsLoss(pos_weight=...)`

This increases the penalty for misclassifying rare diseases.

9. Model Training

Optimizer

Stochastic Gradient Descent (SGD)

Learning Rate

0.01

Momentum

0.9

Weight Decay

0.001

Learning Rate Scheduler

`ReduceLROnPlateau`

Epochs

10

The best model during training was automatically saved.

Screenshot 5

Insert Training Log Screenshot.

10. Evaluation Metrics

The following evaluation metrics were used:

- AUROC
- PR-AUC
- Precision

- Recall
- Macro F1 Score
- Micro F1 Score

These metrics provide a comprehensive assessment of model performance.

11. Results

CustomCNN Results

Mean AUROC: 0.6134

Mean PR-AUC: 0.0736

Macro F1: 0.1240

Micro F1: 0.1543

Screenshot 6

Insert CustomCNN Evaluation Output.

ResNet18 Results

Mean AUROC: 0.5852

Mean PR-AUC: 0.0653

Macro F1: 0.1111

Micro F1: 0.1483

Screenshot 7

Insert ResNet18 Evaluation Output.

VGG19 Results

Mean AUROC: 0.6395

Mean PR-AUC: 0.0810

Macro F1: 0.1326

Micro F1: 0.1432

Screenshot 8

Insert VGG19 Evaluation Output.

12. Model Comparison

Model	Mean AUROC	Mean PR-AUC	Macro F1	Micro F1
CustomCNN	0.6134	0.0736	0.1240	0.1543
ResNet18	0.5852	0.0653	0.1111	0.1483
VGG19	0.6395	0.0810	0.1326	0.1432

Observation

Among all evaluated models, VGG19 achieved the highest AUROC and PR-AUC scores, indicating superior classification performance.

Screenshot 9

Insert Model Comparison Table.

13. Explainable AI using Grad-CAM

Deep learning models often function as black boxes.

To improve transparency, Grad-CAM was implemented.

Purpose

- Visualize model attention
- Explain predictions
- Improve clinician trust

The Grad-CAM heatmap highlights image regions that contributed most to the prediction.

Screenshot 10

Insert CustomCNN Grad-CAM Output.

Screenshot 11

Insert ResNet18 Grad-CAM Output.

Screenshot 12

Insert VGG19 Grad-CAM Output.

14. Retrieval-Augmented Generation (RAG)

A medical knowledge base was developed to provide clinical context for predictions.

Example

Prediction:

Effusion

Retrieved Knowledge:

Pleural effusion is the accumulation of excess fluid within the pleural space surrounding the lungs.

Benefits

- Improves explainability
 - Provides clinical interpretation
 - Supports physician decision-making
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Screenshot 13

Insert AI Clinical Summary Output.

15. Kaggle Execution Environment

The NIH dataset exceeds 45 GB in size.

Due to its large size, Kaggle was selected as the execution platform.

Configuration

- GPU: NVIDIA T4
- Internet: Disabled
- Runtime: Kaggle Notebook

Execution Steps

1. Login to Kaggle.
 2. Create a new notebook.
 3. Import the project notebook.
 4. Enable GPU acceleration.
 5. Disable internet access.
 6. Save and Run All cells.
 7. Download generated model files and outputs.
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Screenshot 14

Insert Kaggle GPU Configuration Screenshot.

Screenshot 15

Insert Save & Run All Screenshot.

16. Conclusion

This project successfully developed an AI-assisted chest disease detection system using deep learning, Explainable AI, and Retrieval-Augmented Generation.

Key Achievements

- Multi-label disease classification
- Patient-wise leakage-safe split
- Comparison of three deep learning architectures
- Grad-CAM-based explainability
- Clinical summary generation using RAG
- Comprehensive performance evaluation

Best Performing Model

VGG19

Mean AUROC: 0.6395

Mean PR-AUC: 0.0810

The integration of deep learning, explainability, and clinical knowledge demonstrates the potential of AI-assisted systems in supporting radiological diagnosis.

17. Future Enhancements

- FastAPI deployment
 - Real-time inference API
 - PACS integration
 - LLM-powered radiology reports
 - Additional explainability methods
 - Clinical validation using external datasets
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Disclaimer

This project is intended for research and educational purposes only. It is not a medical device and should not replace professional medical judgment or diagnosis.